

TAIAROA HEAD, New Zealand

WMO: 938960

Lat: 45.775S

Lon: 170.727E

Elev: 249

StdP: 14.56

Time Zone: 12.00 (NZL)

Period: 94-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 36.8	(c) 38.2	(d) 24.7	(e) 19.0	(f) 45.3	(g) 27.1	(h) 21.2	(i) 44.3	(j) 51.9	(k) 43.2	(l) 48.1	(m) 41.6	(n) 22.9	(o) 240	(p) 0.805

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.1	69.4	58.1	66.6	56.8	64.3	55.9	61.2	65.6	59.8	63.5	58.7	62.0	17.7	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.4	76.2	62.7	58.2	73.1	61.3	57.1	70.2	60.1	27.4	66.0	26.4	63.6	25.7	62.3	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 52.5	(b) 46.7	(c) 40.9	(e) 33.6	(f) 78.4	(g) 1.6	(h) 4.3	(i) 32.4	(j) 81.5	(k) 31.5	(l) 84.1	(m) 30.6	(n) 86.5	(o) 29.4	(p) 89.6		
(a) 52.5	(b) 46.7	(c) 40.9	(e) 31.8	(f) 64.7	(g) 0.8	(h) 2.9	(i) 31.1	(j) 66.7	(k) 30.7	(l) 68.4	(m) 30.2	(n) 70.0	(o) 29.6	(p) 72.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	56.8	57.8	55.7	53.0	50.3	46.3	44.6	46.3	48.0	50.5	51.5	55.0
	DBStd	5.69	3.74	3.94	3.55	3.38	4.34	3.68	3.32	3.81	4.29	4.02	3.99	3.63
	HDD50	639	1	1	1	11	47	121	171	128	89	40	27	2
	HDD65	5017	255	203	291	361	457	561	631	580	510	450	406	312
	CDD50	1103	213	221	176	100	55	10	5	13	29	55	70	156
	CDD65	5	1	2	1	0	0	0	0	0	0	0	0	1
	CDH74	21	1	7	6	0	0	0	0	0	0	0	0	7
	CDH80	2	0	0	0	0	0	0	0	0	0	0	0	2
(14) Wind	WSAvg	16.5	15.8	14.4	16.3	16.3	16.6	16.6	15.7	17.0	16.6	17.5	18.2	16.6
(15) Precipitation	PrecAvg	32.9	3.1	3.0	2.6	2.5	3.1	2.4	2.5	2.8	2.3	2.7	2.6	3.2
	PrecMax	40.8	6.7	8.1	6.8	6.9	11.5	4.9	6.7	6.9	4.1	5.7	5.2	5.0
	PrecMin	25.2	0.8	0.9	0.7	0.9	0.8	1.0	1.0	0.7	0.9	0.9	1.0	1.4
	PrecStd	4.3	1.6	2.0	1.3	1.3	2.4	1.1	1.5	1.7	0.9	1.1	1.0	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.5	73.8	73.3	66.7	66.7	59.2	59.7	60.3	64.5	67.6	68.7	70.5
		MCWB	59.6	58.6	61.9	54.4	53.2	48.9	47.6	48.0	50.1	52.8	55.6	57.0
	2%	DB	67.3	68.8	67.6	63.0	63.0	56.6	53.9	57.3	60.6	62.8	63.3	65.4
		MCWB	58.8	59.3	57.2	53.3	51.6	47.7	45.2	47.2	48.8	51.0	52.7	55.6
	5%	DB	64.5	65.5	63.0	60.0	57.8	53.6	50.9	54.4	56.9	59.1	59.9	62.2
		MCWB	58.5	58.9	56.3	52.8	50.8	46.7	45.2	46.2	48.4	51.4	53.4	54.9
	10%	DB	62.1	62.9	60.4	57.6	55.2	51.5	49.1	51.5	54.1	56.2	57.2	60.0
		MCWB	57.8	58.1	55.7	52.5	50.0	46.2	44.6	45.3	47.7	50.3	52.1	54.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.8	63.6	62.0	57.3	54.8	50.3	48.8	50.1	52.2	55.6	58.1	60.2
		MCDB	66.9	68.4	71.1	61.7	60.4	55.0	52.8	55.3	59.1	62.4	63.5	63.6
	2%	WB	60.8	61.6	59.3	55.6	53.5	49.1	47.5	48.6	50.6	53.6	55.7	58.4
		MCDB	64.7	65.5	63.8	58.6	57.1	53.4	51.5	53.5	56.6	59.1	59.7	62.5
	5%	WB	59.5	60.0	58.0	54.4	52.5	48.1	46.5	47.6	49.5	52.3	54.3	57.2
		MCDB	63.0	63.2	61.1	57.4	56.1	52.3	49.5	52.0	54.8	56.9	57.7	60.8
	10%	WB	58.4	58.9	56.8	53.4	51.3	46.9	45.5	46.6	48.4	51.1	52.8	55.6
		MCDB	61.5	61.9	59.5	56.5	54.8	50.7	48.4	50.4	52.8	55.0	56.2	58.5
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.1	7.9	7.2	7.0	6.4	5.9	7.0	7.8	7.9	8.3	8.0
		MCDBR	11.9	12.6	12.6	11.4	11.8	9.8	8.8	11.6	13.0	12.8	13.1	12.5
		MCWBR	7.6	6.5	6.8	6.7	6.1	6.0	5.7	6.3	6.8	6.8	8.0	7.9
	5% WB	MCDBR	9.8	9.6	10.4	8.5	8.1	6.3	8.2	10.5	9.8	9.8	9.8	10.4
		MCWBR	7.1	6.0	6.8	6.1	5.1	5.8	5.3	5.4	6.6	6.3	7.1	7.3
(40) Clear-Sky Solar Irradiance	taub		0.332	0.330	0.312	0.299	0.287	0.282	0.278	0.297	0.310	0.319	0.320	0.328
	taud		2.575	2.584	2.639	2.653	2.641	2.606	2.630	2.606	2.561	2.558	2.580	2.589
	Ebn at Noon		313	305	296	276	254	241	255	272	291	305	316	317
	Edh at Noon		32	30	26	21	17	16	17	22	27	30	32	32
(44) All-Sky Solar Radiation	RadAvg		1845	1641	1222	785	479	365	446	673	1065	1495	1822	1951
	RadStd		149	133	53	54	29	25	29	41	57	79	79	115

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	+1.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page